

Multi-Camera Images

16 Registers / Camera

64 Register Tokens

DINOv2 ViT + LoRA

Patch Tokens

(a)

**Semantic Token Compressor (TC)**

Scene-Level Router
Soft 6-Expert Mix

1 2 3 4 5 6

**6 SwiGLU Experts**
all active

64 Scene Tokens

64 Queries

Ego + Cmd

Trajectory Generator

64 Trajectory Proposals

Dual-Channel CCS
Scene + Patch Gate

Selected Trajectory



Argmax over 64

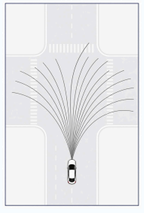
Detached Patches
12×21 / View
1,008 pooled tokens

shared weights

## Training Phase

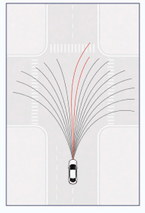
(b)

**Control-Space Augmentation (CSA) - Training Only**



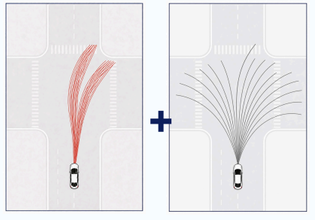
64 Generator Proposals

PDM Seed Ranking



Top 16 Seeds

Control Perturbation + Bicycle Rollout
validity + seed fallback



192 CSA slots + 64 original proposals

Shared CCS

Aux. Score Loss (PDM targets)